THE N-SKELETON OF A SIMPLICIAL SET AS A SUBOBJECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: all three cited lemmas are registered by this batch (L46284, L46287, L46282); the final 'result follows easily' is the unroll: the image contains all nondegenerate simplices of degree at most n and is a subobject, hence equals U'.

Editorial note. The label lemma-n-skeleton-sets names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Nondegenerate simplices of i_{n!}. Same unit, Lemma lemma-imshriek-sets. Lemma lemma-imshriek-sets provides that i_{n!} sk_n U has nondegenerate simplices only in degrees at most n.
- Generated subobject U'. Same unit, Lemma lemma-simplicial-set-n-skel-sub. The target subobject U' is the one constructed in Lemma lemma-simplicial-set-n-skel-sub.
- Injectivity criterion. Same unit, Lemma lemma-injective-map-simplicial-sets. The map is proved injective by Lemma lemma-injective-map-simplicial-sets: injective on nondegenerate simplices with nondegenerate images.

Source reference. Original reference 018O: sources/stacks-simplicial/pinned-original-source.tex, source lines 3233–3233. Prerequisite pointer invoked at proof line(s) 3261; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 017S: sources/stacks-simplicial/pinned-original-source.tex, source lines 2041–2041. Prerequisite pointer invoked at proof line(s) 3269; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 017T: sources/stacks-simplicial/pinned-original-source.tex, source lines 2074–2074. Prerequisite pointer invoked at proof line(s) 3257; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let U be a simplicial set. Let $n \geq 0$ be an integer. The morphism $i_{n!}sk_nU \to U$ identifies $i_{n!}sk_nU$ with the simplicial set $U' \subset U$ defined in Lemma 017T.*

Proof. By Lemma 018O the only nondegenerate simplices of $i_{n!}\mathrm{sk}_nU$ are in degrees $\leq n$. The map $i_{n!}\mathrm{sk}_nU \to U$ is an isomorphism in degrees $\leq n$. Combined we conclude that the map $i_{n!}\mathrm{sk}_nU \to U$ maps nondegenerate simplices to nondegenerate simplices and no two nondegenerate simplices have the same image. Hence Lemma 017S applies. Thus $i_{n!}\mathrm{sk}_nU \to U$ is injective. The result follows easily from this. □